

**Duration: 3 Hours****Maximum Marks:35**

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Q1. Create a dataframe containing information about different vehicles such as Fuel\_Efficiency, Engine\_Power, Vehicle\_Weight, Engine\_Size, and Acceleration. Add 5 records in dataframe, including outlier. Create Box Plots for Fuel Efficiency and Engine Power and identify outliers.

[10 Marks]

Q2. Write a Python program to create a DataFrame with the attributes Temperature, Humidity, Wind Speed, Pressure, and Rainfall. Apply the following data transformation techniques and display the transformed data:

a)Min-MaxScaling

b)Standardization

c) Normalization

[20 Marks]

**OR**

Q2. Implement the Random Forest Classification algorithm on the given Heart Disease dataset to predict the presence or absence of heart disease. Perform the necessary data preprocessing, model training, and testing, and evaluate the model using Accuracy, Precision, Recall, F1-Score, and Confusion Matrix.

[20 Marks]

Q3. Viva

[5 Marks]